Quantitative Portfolio Optimization

Investment Advisory Model for Private Investors

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Investment Advisory Report

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1 Introduction

This report outlines the investment advisory model proposed for implementation within the Private Investor division of the Bank. The framework is designed to translate client-specific characteristics into rigorously constructed, quantitatively optimised portfolios, while remaining operationally tractable and consistent with standard regulatory requirements for investor profiling.

The model rests on two structural pillars. The first is a *client profiling framework*, which classifies investors along two orthogonal dimensions: financial knowledge and experience on the one hand, and a composite risk-return preference score on the other. This dual classification produces a segmentation of clients into five risk profiles, each associated with a distinct strategic asset allocation. The second pillar is a *portfolio construction engine*, which combines Markowitz mean-variance optimisation with Michaud resampling to produce robust Strategic Asset Allocation (SAA) portfolios. For clients with sufficient financial sophistication — referred to hereafter as *informed clients* — the model further allows the incorporation of forward-looking views from the Bank's Global Market Research team through a Black-Litterman framework, introducing a layer of Partial Tactical Asset Allocation (TAA).

The remainder of this report is structured as follows. Section 2 describes the client profiling methodology. Section 3 defines the asset universe available to each client segment. Section 4 presents the quantitative constraints governing portfolio construction. Section 5 details the mathematical formulation of the optimisation problems.

2 Customer Profiling Framework

2.1 Overview

The profiling framework classifies clients along three dimensions: (i) financial knowledge and experience, (ii) risk tolerance, and (iii) investment profile. Each dimension is assessed through a structured questionnaire, and each produces a normalised score in $[0, 100]$. The combination of these scores determines both the *client segment* (informed vs. non-informed) and the *risk profile* (one of five ordered categories).

2.2 Informed vs. Non-Informed Clients

The primary segmentation distinguishes between *informed* and *non-informed* clients on the basis of a financial knowledge and experience score $S_K \in [0, 100]$. This score is constructed from questionnaire items assessing, the client's familiarity with asset price volatility, the mechanics of financial leverage, the risk-return trade-off in fixed income and equity instruments, and prior direct investment experience.

A client is classified as *informed* if and only if $S_K > 60$; otherwise, the client is classified as *non-informed.*

Rationale. Although this binary split is a simplification of a continuous spectrum of financial literacy, it captures a meaningful and practically relevant form of investor heterogeneity. Informed clients possess the conceptual vocabulary to understand portfolio volatility, drawdown dynamics, and the role of diversification. This awareness enables them to tolerate short-term losses without abandoning their investment mandate, and to evaluate the implications of tactical deviations from a strategic benchmark. Non-informed clients, by contrast, may misinterpret market fluctuations as signals of model failure, leading to behavioural responses, such as premature liquidation, that are detrimental to long-term outcomes. The threshold of 60 is calibrated to ensure that only clients with demonstrated, above-average financial literacy are exposed to the broader asset universe and to research-driven tactical views, whose potential for adverse outcomes must be consciously understood and accepted.

2.3 Risk Tolerance and Investment Profile

Risk Tolerance. Risk tolerance is defined as the maximum degree of financial uncertainty that an investor is both *willing* and *able* to bear over a given investment horizon. It is a composite construct derived from two distinct but complementary components:

- **Risk Attitude** (A): the psychological propensity of the individual to accept financial losses. It is elicited through behavioural questions, including the maximum portfolio drawdown the client would tolerate before liquidating (expressed as a percentage of wealth, consistent with a Value-at-Risk framing over a one-year horizon).
- **Risk Capacity** (C): the objective financial ability of the individual to sustain losses without compromising their financial wellbeing. It is assessed along four dimensions: (i) *investment horizon*, the most critical factor, as a longer horizon allows recovery from adverse market cycles; (ii) *income stability*, clients with stable, predictable cash flows have greater capacity to withstand portfolio drawdowns; (iii) *net worth and liquidity*, the presence of an emergency fund and liquid assets increases capacity to hold illiquid or volatile positions; and (iv) *financial obligations*, liabilities such as mortgage repayments, education expenses, or other recurring commitments reduce the investable surplus and constrain risk-taking.

The composite risk tolerance score is defined as:

$$S_{RT} = \frac{1}{2}A + \frac{1}{2}C, \qquad S_{RT} \in [0, 100]. \tag{1}$$

Investment Profile. The investment profile score $S_{IP} \in [0, 100]$ captures the client's stated financial objective, ranging from capital preservation (low score) to capital growth (high score). It is elicited through questions about return expectations, the acceptable trade-off between risk and return, and the intended use of the invested capital.

2.4 Final Score and Risk Profile Classification

Once the client segment (informed or non-informed) has been determined, the final classification score $\mathcal{S} \in [-100, 100]$ is computed as:

$$\mathcal{S} = S_{IP} - S_{RT}, \tag{2}$$

where the investment profile enters positively, reflecting that a higher growth orientation supports a more aggressive allocation, and risk tolerance enters negatively, reflecting that a higher risk aversion supports a more conservative allocation. The equally weighted combination ensures that both dimensions contribute symmetrically to the final placement.

The score $\mathcal{S}$ is then mapped to one of five ordered risk profiles according to the following rule:

Risk Profile	**Score Range**
Ultra-Conservative	$\mathcal{S} \in (-100,\ -60)$
Conservative	$\mathcal{S} \in [-60,\ -20)$
Balanced	$\mathcal{S} \in [-20,\ +20]$
Aggressive	$\mathcal{S} \in (+20,\ +60]$
Ultra-Aggressive	$\mathcal{S} \in (+60,\ +100]$

Table 1: Mapping from final score $\mathcal{S}$ to risk profile.

3 Asset Universe

3.1 Base Universe (Non-Informed and Informed Clients)

The base asset universe is composed of eleven instruments spanning the main investable asset classes: fixed income, global equity, and real assets. Table 2 provides a summary.

Ticker	Index	Asset Class	Role
BOND_ST_AGG	Bloomberg Global Agg 1–3Y EUR Unhedged	Fixed Income	Short-duration diversifier
BOND_GOV_GLO	Bloomberg Global Agg Treasuries USD Unhedged	Fixed Income	Sovereign, all durations
BOND_CORP_IG	Bloomberg Global Agg Corporate USD Unhedged	Fixed Income	Investment-grade credit
BOND_HY	Bloomberg Global High Yield USD Unhedged	Fixed Income	High-yield credit
BOND_EM	Bloomberg EM Sovereign USD Unhedged	Fixed Income	Emerging market sovereign
BOND_GOV_EUR	Bloomberg EuroAgg Government EUR Unhedged	Fixed Income	European sovereign (tilt)
EQ_EUR	MSCI Europe Gross TR Local	Equity	European developed markets
EQ_US	MSCI North America Gross TR Local	Equity	North American markets
EQ_PAC	MSCI Pacific Gross TR Local	Equity	Asia-Pacific developed
EQ_EM	MSCI EM Gross TR Local	Equity	Emerging market equity
GOLD	Bloomberg Gold Sub-Index TR	Real Assets	Volatility hedge

Table 2: Base asset universe.

The fixed income segment differentiates instruments along three dimensions: (i) *duration* — short-term (1–3 years) versus all-duration global aggregates; (ii) *geography* — global, European, and emerging market; and (iii) *credit quality* — investment-grade sovereign, investment-grade corporate, and high-yield corporate. This granularity allows the optimisation to select the appropriate risk-return positioning within the fixed income allocation for each profile.

The equity segment is represented by four cap-weighted regional indices covering Europe, North America, Asia-Pacific, and Emerging Markets. These indices serve as the core building blocks for geographic diversification and are expressed in local currency terms to isolate equity market returns from currency effects within the optimisation.

Gold is included as a real asset diversifier. Its historically low or negative correlation with equities during periods of market stress provides a stabilising effect on portfolio volatility, particularly for conservative profiles.

The Bloomberg EuroAgg Government index (`BOND_GOV_EUR`) is included in the base universe to accommodate clients exhibiting *home country bias*. It is inactive by default (lower bound of zero) and is activated only through client-specific overlay constraints, together with an increased allocation to `EQ_EUR`.

3.2 Extended Universe (Informed Clients Only)

In addition to the base universe, informed clients may access eighteen global sectoral equity indices and a broad commodity aggregate:

- **Sectoral equity indices**: MSCI World sector indices covering Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials (Europe), Information Technology, Communication Services, Utilities, Media & Entertainment, Pharmaceuticals, Life Sciences, Banks, Biotechnology, Chemicals, Real Estate, and Software.
- **Commodity aggregate**: Bloomberg Commodity Total Return Index (`COMMOD_AGG`).

These instruments are inactive by default and are activated exclusively in two circumstances: (i) when the client wishes to express a sector or commodity tilt as part of a client-specific overlay, subject to a maximum individual allocation of 15%; or (ii) when the Bank's Global Market Research team issues a view on a specific sector or commodity, in which case the Black-Litterman framework unlocks the relevant instrument and incorporates the view into the posterior expected return vector.

The rationale for restricting this universe to informed clients is twofold. First, sectoral and commodity indices exhibit higher idiosyncratic risk and lower diversification relative to broad market benchmarks; their inclusion requires the ability to evaluate concentration risk. Second, research-driven views are inherently uncertain; if incorrect, they may reduce portfolio performance, a consequence that informed clients are better equipped to understand and accept ex ante.

4 Quantitative Constraints

4.1 Macro Asset-Class Bounds

Each risk profile is subject to a set of macro-level inequality constraints that define admissible ranges for aggregate equity, aggregate bond, and aggregate commodity exposure. Let w_E, w_B, and w_C denote the total portfolio weight allocated to equities, bonds, and commodities, respectively. The constraints take the form:

$$w_E^{\min}(h) \;\le\; w_E \;\le\; w_E^{\max}(h), \qquad w_C^{\min}(h) \;\le\; w_C \;\le\; w_C^{\max}(h), \tag{3}$$

with the residual allocated to fixed income: $w_B = 1 - w_E - w_C$. The profile-specific bounds are reported in Table 3.

Profile	$w_E^{\min}$	$w_E^{\max}$	$w_B^{\min}$	$w_B^{\max}$	$w_C^{\min}$	$w_C^{\max}$
Ultra-Conservative	5%	15%	75%	95%	0%	10%
Conservative	15%	35%	55%	85%	0%	10%
Balanced	35%	50%	40%	65%	0%	10%
Aggressive	50%	75%	13%	50%	0%	12%
Ultra-Aggressive	75%	100%	0%	25%	0%	12%

Table 3: Macro asset-class bounds by risk profile. Bond bounds are implied by $w_B = 1 - w_E - w_C$.

Rationale. The macro constraints encode the fundamental risk-return trade-off between asset classes. Equities offer higher long-run expected returns but are characterised by greater volatility, fat-tailed return distributions, and larger drawdowns. Fixed income, particularly high-quality sovereign bonds, provides lower volatility, income stability, and a degree of flight-to-quality hedging during equity market stress. Commodities, represented primarily by gold, contribute diversification benefits through low correlation with both equities and bonds.

The monotone increase in equity bounds across profiles reflects the principle that investors with higher risk tolerance and longer investment horizons should hold more equity, thereby accepting short-term volatility in exchange for higher expected terminal wealth. Conversely, conservative profiles maintain a predominantly fixed income allocation to limit drawdown risk and preserve capital. The commodity upper bound is kept moderate across all profiles (at most 12%) to prevent excessive concentration in a single real asset, while still allowing gold to play its intended diversification role.

4.2 Relative Intra-Asset-Class Bounds

Within each asset class bucket, relative bounds are imposed to control the composition of fixed income and equity allocations. These are expressed as lower and upper limits on the weight of each individual instrument *relative to the total bucket weight*. Formally, for instrument i belonging to bucket $\mathcal{B}$ with total weight $W_{\mathcal{B}} = \sum_{j \in \mathcal{B}} w_j$:

$$r_i^{\min}(h) \cdot W_{\mathcal{B}} \;\le\; w_i \;\le\; r_i^{\max}(h) \cdot W_{\mathcal{B}}. \tag{4}$$

The profile-specific relative bounds are reported in Table 4.

Asset	Ticker	UC	C	B	A	UA
Fixed Income (relative to total bond weight)						
Short-term agg.	BOND_ST_AGG	[0, 0.60]	[0, 0.60]	[0, 0.60]	[0, 0.60]	[0, 0.60]
Global sovereign	BOND_GOV_GLO	[0, 0.50]	[0, 0.50]	[0, 0.50]	[0, 0.50]	[0.05, 0.50]
Corp. invest. grade	BOND_CORP_IG	[0, 0.25]	[0, 0.25]	[0.05, 0.40]	[0.05, 0.50]	[0.10, 0.40]
High yield	BOND_HY	[0, 0.05]	[0, 0.05]	[0, 0.10]	[0, 0.30]	[0, 0.30]
Emerging sovereign	BOND_EM	[0, 0.10]	[0, 0.10]	[0, 0.20]	[0, 0.30]	[0, 0.30]
European sovereign	BOND_GOV_EUR	[0, 0]	[0, 0]	[0, 0]	[0, 0]	[0, 0]
Equity (relative to total equity weight)						
Europe	EQ_EUR	[0.10, 0.50]	[0.10, 0.50]	[0.10, 0.50]	[0.10, 0.50]	[0.10, 0.50]
North America	EQ_US	[0.40, 0.60]	[0.40, 0.60]	[0.40, 0.60]	[0.40, 0.60]	[0.30, 0.60]
Asia-Pacific	EQ_PAC	[0, 0.15]	[0, 0.15]	[0, 0.15]	[0, 0.20]	[0, 0.20]
Emerging markets	EQ_EM	[0, 0.15]	[0, 0.15]	[0, 0.15]	[0, 0.20]	[0, 0.20]

Table 4: Relative intra-asset-class bounds by risk profile. UC = Ultra-Conservative, C = Conservative, B = Balanced, A = Aggressive, UA = Ultra-Aggressive.

Rationale. The relative bounds serve three purposes. First, they enforce a degree of *intra-class market neutrality*: by imposing non-trivial lower bounds on major instruments (e.g., a minimum 40% North America equity weight), they prevent extreme concentration in a single geography or credit segment, ensuring that each portfolio remains broadly representative of the global opportunity set. Second, they allow for *profile-consistent tilts toward higher-risk, higher-return instruments*: more aggressive profiles permit larger allocations to high-yield bonds, emerging market debt, and emerging market equity, instruments that carry higher credit or liquidity risk but offer commensurately higher expected returns. Third, the `BOND_GOV_EUR` bound is fixed at zero by default for all profiles, ensuring that European sovereign overweighting is activated only through the explicit client-level request mechanism.

4.3 Client-Specific Overlay Constraints

In addition to the macro and relative constraints described above, clients may request up to two additional *overlay constraints* reflecting personal preferences, subject to the following limits:

- Any overlay on `BOND_GOV_EUR` or `EQ_EUR` (home country bias): minimum 5%, maximum 20%.
- Any overlay on sectoral equity indices (informed clients only): maximum individual allocation of 15%.

These client-specific constraints are implemented as additional box constraints within the optimisation problem and are layered on top of the standard macro and relative bounds.

5 Mathematical Formulation

5.1 Strategic Asset Allocation: Markowitz Optimisation with Michaud Resampling

5.1.1 Base Optimisation Problem

For each risk profile $h \in \{\text{UC, C, B, A, UA}\}$, the Strategic Asset Allocation is obtained by solving a constrained minimum-variance problem. Let $\mathbf{w} \in \mathbb{R}^n$ denote the vector of portfolio weights, $\boldsymbol{\mu} \in \mathbb{R}^n$ the vector of annualised expected returns, and $\boldsymbol{\Sigma} \in \mathbb{R}^{n\times n}$ the annualised covariance matrix of asset returns. The optimisation problem is:

$$\min_{\mathbf{w}} \quad \frac{1}{2}\mathbf{w}^\top \boldsymbol{\Sigma}\,\mathbf{w} \tag{5}$$

subject to:

$$\sum_{i=1}^{n} w_i = 1 \tag{6}$$

$$w_i \geq 0 \quad \forall\, i \tag{7}$$

$$\mathbf{w}^\top \boldsymbol{\mu} \geq \mu_T(h) \tag{8}$$

$$w_E^{\min}(h) \leq \sum_{i\in\mathcal{E}} w_i \leq w_E^{\max}(h) \tag{9}$$

$$w_C^{\min}(h) \leq \sum_{i\in\mathcal{C}} w_i \leq w_C^{\max}(h) \tag{10}$$

$$r_i^{\min}(h)\cdot W_{\mathcal{B}} \leq w_i \leq r_i^{\max}(h)\cdot W_{\mathcal{B}} \quad \forall\, i\in\mathcal{B},\ \mathcal{B}\in\{\mathcal{E},\mathcal{F}\} \tag{11}$$

where $\mathcal{E}$, $\mathcal{F}$, $\mathcal{C}$ denote the sets of equity, fixed income, and commodity instruments, respectively, and $W_{\mathcal{B}} = \sum_{j\in\mathcal{B}} w_j$ is the total bucket weight.

Constraint (8) requires the portfolio to achieve at least the target return $\mu_T(h)$ associated with profile h. The target returns, expressed on an annualised basis, are set as follows:

Risk Profile	$\mu_T(h)$
Ultra-Conservative	3.3%
Conservative	4.8%
Balanced	6.3%
Aggressive	8.3%
Ultra-Aggressive	10.3%

Table 5: Annualised target returns by risk profile.

These targets represent the minimum acceptable expected return for each profile and are calibrated to be consistent with historical long-run asset class premia. The objective function minimises portfolio variance subject to these constraints, yielding the minimum-variance portfolio on the efficient frontier that satisfies the client's return requirement.

5.1.2 Michaud Resampling

A well-known limitation of classical Markowitz optimisation is its sensitivity to estimation error in $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$: small perturbations in the inputs can produce large and unstable changes in optimal weights. To mitigate this, the framework employs the *Michaud resampling* procedure, which averages optimal portfolios obtained from multiple simulated versions of the input parameters.

Let $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ denote the sample estimates of expected returns and the covariance matrix, computed from T historical observations. The resampling procedure proceeds as follows:

1. For $s = 1,\ldots,S$ (with $S = 100$ in this implementation), draw a simulated expected return vector:
$$\tilde{\boldsymbol{\mu}}^{(s)} \sim \mathcal{N}\left(\hat{\boldsymbol{\mu}},\ \frac{\hat{\boldsymbol{\Sigma}}}{T}\right). \tag{12}$$

2. Solve the constrained optimisation problem (5)–(11) with $\boldsymbol{\mu} = \tilde{\boldsymbol{\mu}}^{(s)}$ and $\boldsymbol{\Sigma} = \hat{\boldsymbol{\Sigma}}$, obtaining the optimal weight vector $\mathbf{w}^{(s)}$.

3. Compute the resampled portfolio as the arithmetic average of the simulated optima:

$$\mathbf{w}^* = \frac{1}{S}\sum_{s=1}^{S}\mathbf{w}^{(s)}. \tag{13}$$

By averaging over S draws from the sampling distribution of $\hat{\boldsymbol{\mu}}$, the procedure implicitly accounts for parameter uncertainty and produces portfolios that are more stable and diversified than their single-optimisation counterparts. The covariance matrix $\hat{\boldsymbol{\Sigma}}$ is held fixed across draws, as its estimation is less sensitive to sample variation than the mean vector. Each individual optimisation in step 2 is solved using Sequential Quadratic Programming (SLSQP) with $R = 7$ random restarts to mitigate the risk of convergence to local minima, retaining the solution with the lowest realised volatility.

5.2 Partial Tactical Asset Allocation: Black-Litterman Framework (Informed Clients Only)

For informed clients who wish to incorporate the views of the Bank's Global Market Research (GMR) team, the Strategic Asset Allocation weights are updated on a semi-annual basis using the *Black-Litterman* (BL) model. This introduces a layer of Partial Tactical Asset Allocation, whereby the posterior expected return vector is tilted toward the GMR views while remaining anchored to the strategic equilibrium.

5.2.1 Equilibrium Prior Returns

The Black-Litterman model begins from a set of *equilibrium implied returns* $\boldsymbol{\pi}$, derived by reverse-engineering the expected returns consistent with the SAA portfolio weights $\mathbf{w}^*$ under the assumption that this portfolio is mean-variance optimal:

$$\boldsymbol{\pi} = \lambda\,\boldsymbol{\Sigma}\,\mathbf{w}^*, \tag{14}$$

where $\lambda > 0$ is the risk-aversion coefficient, calibrated as:

$$\lambda = \frac{\mu_T(h)}{(\mathbf{w}^*)^\top\boldsymbol{\Sigma}\,\mathbf{w}^*}. \tag{15}$$

This ensures that the implied equilibrium return is consistent with the target return of the respective risk profile.

5.2.2 View Specification

GMR views are expressed as K linear statements on expected returns. Let $\mathbf{P} \in \mathbb{R}^{K\times n}$ be the *pick matrix*, where each row identifies the assets involved in a given view, and $\mathbf{q} \in \mathbb{R}^K$ be the *view vector* of expected returns. Two types of views are supported:

- **Absolute view**: the expected return of asset i equals q_k. The corresponding row of $\mathbf{P}$ has a single entry of $+1$ in position i.
- **Relative view**: asset i is expected to outperform asset j by q_k. The corresponding row of $\mathbf{P}$ has $+1$ in position i and -1 in position j.

5.2.3 View Uncertainty

The uncertainty associated with each view is encoded in the diagonal matrix $\boldsymbol{\Omega} \in \mathbb{R}^{K \times K}$, whose k-th diagonal element is:

$$\Omega_{kk} = \delta_k \cdot \tau \, \mathbf{p}_k^\top \boldsymbol{\Sigma} \, \mathbf{p}_k, \tag{16}$$

where $\mathbf{p}_k$ is the k-th row of $\mathbf{P}$, $\tau = 1/T$ is a scaling parameter reflecting the uncertainty in the prior (with T denoting the number of observations used to estimate $\boldsymbol{\Sigma}$), and $\delta_k \in \{0.5, 1.0, 2.0\}$ is a confidence scalar assigned by the analyst:

Confidence Level	δ_k	**Interpretation**
High	0.5	View twice as precise as the prior
Medium	1.0	View and prior equally uncertain (He-Litterman)
Low	2.0	View half as precise as the prior

Table 6: Confidence scalar δ_k and its interpretation.

5.2.4 Posterior Expected Returns

The Black-Litterman posterior expected return vector $\boldsymbol{\mu}_{\mathrm{BL}}$ is obtained by combining the prior $\boldsymbol{\pi}$ with the GMR views via Bayes' theorem:

$$\boldsymbol{\mu}_{\mathrm{BL}} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{P}\right]^{-1} \left[(\tau\boldsymbol{\Sigma})^{-1}\boldsymbol{\pi} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{q}\right]. \tag{17}$$

The posterior covariance matrix, which incorporates both market uncertainty and view uncertainty, is:

$$\boldsymbol{\Sigma}_{\mathrm{BL}} = \boldsymbol{\Sigma} + \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{P}\right]^{-1}. \tag{18}$$

5.2.5 Constrained Optimisation after Black-Litterman

Given the posterior moments ($\boldsymbol{\mu}_{\mathrm{BL}}$, $\boldsymbol{\Sigma}_{\mathrm{BL}}$), the optimal BL portfolio is obtained by solving the same constrained minimum-variance problem as in (5)–(11), with $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ replaced by their posterior counterparts:

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma}_{\mathrm{BL}} \, \mathbf{w} \tag{19}$$

subject to constraints (6)–(11), with the additional modification that assets mentioned in the GMR views may have their upper bounds expanded to permit tactical overweighting, while assets subject to bearish views are constrained by the unchanged upper bound and the optimiser reduces their allocation endogenously. The problem is solved using SLSQP with $R = 7$ random restarts.

The BL portfolio weights are recomputed semi-annually as new GMR views become available, while the SAA prior $\mathbf{w}^*$ and the covariance matrix $\hat{\boldsymbol{\Sigma}}$ are held fixed across all semi-annual updates. An alternative implementation could employ an expanding or rolling window to update $\hat{\boldsymbol{\Sigma}}$ at each rebalancing date, potentially improving the model's responsiveness to structural breaks and regime changes in asset return dynamics. However, in the interest of consistency with the SAA framework, where parameters are estimated once on the training sample and held fixed throughout the test period, this approach was not adopted here.

6 Results and Performance Analysis

6.1 Backtesting Methodology

The backtesting framework is designed to evaluate the out-of-sample performance of both the Strategic Asset Allocation (SAA) portfolios, constructed via Markowitz mean-variance optimisation with Michaud resampling, and the Partial Tactical Asset Allocation (TAA) portfolios, constructed via the Black-Litterman model. The two evaluations share a common data sample but differ in the asset universe, the test horizon, and the portfolio updating mechanism.

6.1.1 Data and Sample Split

The full dataset consists of monthly total-return observations for the instruments in the base and extended universes described in Sections 3.1–3.2. The sample is partitioned into two non-overlapping sub-periods at observation $T_0 = 200$:

- **In-sample (training) period**: observations $1, \ldots, 200$, used to estimate the annualised expected return vector $\hat{\boldsymbol{\mu}}$ and the annualised covariance matrix $\hat{\boldsymbol{\Sigma}}$ that feed both the SAA optimisation and the Black-Litterman equilibrium prior.
- **Out-of-sample (test) period**: the subsequent 76 monthly observations for the SAA evaluation (non-informed client universe, 11 base assets), and the subsequent 38 monthly observations for the TAA evaluation (informed client universe, up to 30 assets) The variation in duration is due to the differing historical depth of the datasets for the two asset universes (11 vs. 30 assets).

In both cases, the input parameters $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ are estimated once on the training period and held fixed throughout the test period. No rolling re-estimation is performed, so the backtest isolates the out-of-sample stability of the optimised weights rather than the forecasting performance of the estimator.

6.1.2 SAA Backtest: Static Resampled Portfolios

For each of the five risk profiles $h \in \{\text{UC, C, B, A, UA}\}$, the resampled SAA weight vector $\mathbf{w}^*(h)$ is computed once on the in-sample period following the Michaud procedure described in Section 5.1.2. These weights are then applied statically to the out-of-sample return sequence.

Rebalancing scheme. Portfolio weights are reviewed every six months ($\Delta t = 6$ periods). At each scheduled review date t, the realised weight vector $\tilde{\mathbf{w}}_t$, obtained by letting the SAA weights drift with cumulative asset returns since the last rebalancing, is compared with the target $\mathbf{w}^*(h)$. Rebalancing is executed if and only if the ℓ_1 drift exceeds a tolerance threshold $\varepsilon = 0.01$:

$$\sum_{i=1}^{n} \left| \tilde{w}_{t,i} - w_i^*(h) \right| > \varepsilon. \tag{20}$$

If the condition is met, the portfolio is reset to $\mathbf{w}^*(h)$ and transaction costs are applied (see below). If the drift is below the threshold, no trade is executed and no cost is incurred.

Transaction costs. Each rebalancing event incurs both a variable and a fixed cost component. The variable cost is proportional to the total turnover, computed as the ℓ_1 distance between the pre- and post-rebalancing weight vectors, and applied at a rate of $c_v = 10$ basis points on the traded notional. A fixed cost of $c_f = £10$ per traded instrument is also charged. Both cost components are deducted from the portfolio AUM prior to resetting the weights.

Weight dynamics. Between rebalancing dates, portfolio value and weights evolve according to the realised simple returns of the constituent assets. At each period t, the portfolio return is computed as the weighted average of asset returns, and weights are updated by standard normalisation to reflect the change in relative asset values.

6.1.3 TAA Backtest: Black-Litterman with Semi-Annual View Updates

For informed clients, the Black-Litterman framework is applied over the 38-month out-of-sample period. Unlike the SAA backtest, the TAA backtest incorporates time-varying analyst views that are updated at each semi-annual rebalancing date, causing the posterior expected return vector $\boldsymbol{\mu}_{\mathrm{BL}}$ — and consequently the optimal portfolio weights — to change over time.

View schedule and construction. Analyst views are specified at seven semi-annual rebalancing dates ($t = 0, 6, 12, 18, 24, 30, 36$), corresponding to the calendar windows reported in Table 7. At each date, the Global Market Research team issues two views, expressed either as absolute return forecasts for a single asset or as relative outperformance forecasts between two assets, following the pick-matrix formulation described in Section 5.2.2. Each view is assigned one of three confidence levels (high, medium, or low), which governs the diagonal elements of the uncertainty matrix $\boldsymbol{\Omega}$ as specified in Section 5.2.3.

t	**Approx. Date**	**View 1**	**View 2**
0	2023-01	Absolute: `SEC_IT` at 30% (high)	Relative: `EQ_US` over `EQ_EUR` by 6% (mediu
6	2023-07	Relative: `SEC_IT` over `SEC_STAPLES` by 14% (high)	Absolute: `GOLD` at 10% (medium)
12	2024-01	Absolute: `EQ_US` at 20% (medium)	Absolute: `BOND_ST_AGG` at 5% (high)
18	2024-07	Relative: `SEC_FIN` over `SEC_IT` by 8% (medium)	Absolute: `GOLD` at 16% (high)
24	2025-01	Absolute: `SEC_ENERGY` at 24% (high)	Relative: `EQ_US` over `EQ_EM` by 12% (mediu
30	2025-07	Absolute: `SEC_HEALTH` at 14% (medium)	Relative: `BOND_CORP_IG` over `BOND_HY` by 4
36	2026-01	Absolute: `SEC_SOFT` at 20% (high)	Absolute: `SEC_RE` at 10% (medium)

Table 7: Semi-annual GMR view schedule used in the TAA backtest. All return figures are annualised. Confidence levels are in parentheses.

It must be emphasised that these views are selected *ex post*, having been constructed with knowledge of the realised market outcomes over the test period. They are therefore illustrative of the model's *capacity* to incorporate research views and are not intended to represent a genuine real-time forecasting exercise. Any performance attribution that accrues to the TAA layer relative to the SAA benchmark should accordingly be interpreted as an upper bound on the value of accurate GMR views, rather than as an estimate of expected out-of-sample alpha.

Portfolio construction at each rebalancing date. At each semi-annual date t, the Black-Litterman posterior ($\boldsymbol{\mu}_{\mathrm{BL},t}$, $\boldsymbol{\Sigma}_{\mathrm{BL},t}$) is computed using the equilibrium prior derived from the in-sample resampled SAA weights $\mathbf{w}^*$, the fixed covariance matrix $\hat{\boldsymbol{\Sigma}}$, and the active views at date t. The constrained minimum-variance problem described in Section 5.2.5 is then solved using SLSQP with $R = 7$ random restarts, subject to the full set of macro, relative, and client-specific constraints applicable to the informed client universe. If the optimisation fails for a given profile, the framework falls back to the SAA prior weights $\mathbf{w}^*(h)$ for that period, ensuring continuity of the backtest.
Assets referenced in the active views, sectoral indices and commodity aggregates that are inactive by default, are selectively unlocked at each rebalancing date in accordance with the unlock mechanism described in Section 3.2. Their individual allocation is capped at 15% of total portfolio value. At dates where no views are issued, the portfolio defaults to the SAA weights without modification.

Rebalancing scheme and transaction costs. The rebalancing and cost structure for the TAA backtest mirrors that of the SAA backtest: portfolios are reviewed every six months, rebalancing is triggered only when the ℓ_1 drift from the current BL target exceeds the tolerance $\varepsilon = 0.01$, and the same variable ($c_v = 10$ bps) and fixed ($c_f = \pounds 10$ per asset) cost parameters apply. Because the BL target weights change at each semi-annual date, turnover in the TAA backtest is generally higher than in the SAA backtest, reflecting both passive drift and active reallocation driven by view updates.

6.2 Statistics and Consideration

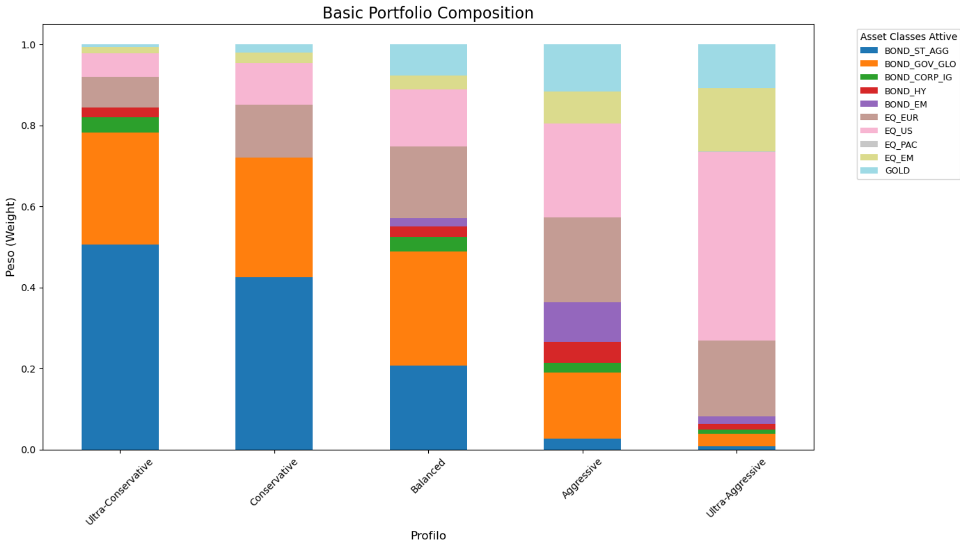

Figure 1: Composition Basic Portfolio (SAA).

Tables 8 reports the SAA weights produced by the Michaud resampling procedure, estimated on the 200-month training period and applied statically throughout the out-of-sample evaluation. These allocations constitute the baseline portfolios for both informed and non-informed clients. It is important to emphasise that in the present simulation neither client type has expressed any additional preference — no home-country bias overlay, no sectoral tilt, and no commodity activation has been requested. The weights therefore reflect the pure output of the constrained optimisation under the default bounds described in Section 4, without any client-specific modification.

Profile	B_ST	B_GOV	B_IG	B_HY	B_EM	B_EUR	EQ_EU	EQ_US	EQ_PAC	EQ_EM	GOLD
Ultra-Conservative	50.65	27.64	3.76	2.36	0.00	0.00	7.50	6.00	0.00	1.50	0.58
Conservative	42.59	29.52	0.00	0.00	0.00	0.00	12.98	10.39	0.00	2.60	1.92
Balanced	20.80	28.04	3.60	2.60	2.08	0.00	17.64	14.21	0.00	3.42	7.61
Aggressive	2.65	16.28	2.41	5.15	9.88	0.00	20.93	23.12	0.00	7.90	11.67
Ultra-Aggressive	0.78	3.17	0.97	1.34	1.98	0.00	18.61	46.69	0.09	15.64	10.75

Table 8: SAA baseline portfolio weights (%) by risk profile, estimated on the 200-month training period. Ticker abbreviations follow Table 2. `BOND_GOV_EUR` is zero across all profiles, consistent with its default inactive status. Values below 10^{-4} are reported as 0.00.

6.2.1 SAA Performance: Risk–Return Profile

The SAA results exhibit a clear and monotone risk–return gradient across profiles. Annualised returns increase from 1.63% for the Ultra-Conservative profile to 12.43% for the Ultra-Aggressive profile, and annualised volatility rises correspondingly from 4.55% to 11.29%. This pattern is consistent with the macro asset-class constraints described in Section 4.1, which progressively shift the allocation toward equity as the profile becomes more aggressive.

Risk-adjusted performance, as measured by the Sharpe ratio, also improves monotonically across profiles, ranging from 0.36 to 1.10. This suggests that, over the test period, bearing additional equity risk was not merely compensated in absolute terms but was also rewarded on a per-unit-of-risk basis. The Calmar ratio follows the same direction, rising from 0.25 to 0.80, though the improvement moderates at the more aggressive end, reflecting the disproportionate increase in maximum drawdown. Maximum drawdown escalates substantially with profile aggressiveness — from −6.57% for Ultra-Conservative to −15.47% for Ultra-Aggressive — and the worst single month widens sharply from −3.14% to −10.60%, indicating that higher-risk profiles are exposed to meaningfully larger short-term losses.

A noteworthy concern emerges when comparing the out-of-sample annualised returns against the profile target returns specified in Table 5. All three conservative profiles fail to meet their respective targets: the Ultra-Conservative portfolio delivers 1.63% against a target of 3.3%, the Conservative portfolio 3.02% against 4.8%, and the Balanced portfolio 5.39% against 6.3%. Realised volatility for these profiles is, at the same time, equal to or slightly above what would be expected from the in-sample optimisation. This combination, lower-than-target returns and unchanged or marginally higher volatility, constitutes a potential structural weakness of the model at the conservative end of the spectrum. The most likely explanation is the well-known sensitivity of mean-variance optimisation to estimation error in $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$: even with Michaud resampling, which averages over simulated perturbations of the expected return vector, the posterior weights remain anchored to in-sample return estimates that may not persist out-of-sample.

6.2.2 TAA Performance: Black-Litterman Portfolio Construction

The TAA results display a structurally different performance profile relative to the SAA, characterised by higher returns, substantially lower volatility, and markedly improved risk-adjusted metrics across all five profiles. The most striking feature is the compression of annualised volatility: across all profiles, TAA volatility lies in the range [4.10%, 6.87%], compared with [4.55%, 11.29%] for SAA. The Ultra-Aggressive profile, in particular, achieves a TAA volatility of 6.87% against a SAA volatility of 11.29%, while simultaneously delivering a higher annualised return (14.63% vs. 12.43%). This combination results in a Sharpe ratio of 2.13 for the TAA Ultra-Aggressive portfolio, compared with 1.10 for its SAA counterpart.

Maximum drawdowns under the TAA framework are strikingly contained and, notably, nearly uniform across profiles, ranging narrowly from −5.54% to −6.58%. This stands in sharp contrast to the SAA drawdowns, which scale from −6.57% to −15.47% with increasing aggressiveness. The Calmar ratio reflects this directly: it rises from 0.64 to 2.22 across TAA profiles, versus 0.25 to 0.80 for SAA. The best-month statistics reveal an additional structural feature: TAA portfolios exhibit considerably more muted upside, ranging from 2.83% to 4.71%, compared with 4.24% to 9.52% under SAA. This is consistent with the view-driven reallocation mechanism, which concentrates risk in assets with strong posterior expected returns rather than allowing broad equity upside to materialise freely, producing a more symmetric return distribution with both tails compressed.

The superior risk-adjusted performance of the TAA portfolios relative to their SAA counterparts can be attributed to two distinct and separable sources. The first is the Bayesian structure of the Black-Litterman estimation itself: by blending the in-sample equilibrium prior $\boldsymbol{\pi} = \lambda\,\boldsymbol{\Sigma}\,\mathbf{w}^*$ with analyst views through the posterior formula (17), the model produces a more stable and regularised estimate of expected returns than the raw sample mean $\hat{\boldsymbol{\mu}}$. The second source is the accuracy of the GMR views incorporated at each semi-annual rebalancing date. As discussed in Section 6.1.3, these views were constructed ex post with knowledge of realised returns; they were therefore systematically correct in identifying assets with above- or below-average performance over each six-month window. The combination of well-calibrated Bayesian estimation and accurate views produces the outperformance. In a live implementation, where views would be formed in real time and subject to forecasting error, the performance contribution of the second source could be substantially diminished, and the net TAA benefit would derive primarily from the regularisation properties of the Black-Litterman framework itself.

6.2.3 Comparative Assessment and Limitations

Taken together, the statistics suggest that the Black-Litterman portfolio construction delivers a substantial improvement in risk-adjusted performance relative to the static SAA across all profiles and all metrics considered. However, several important caveats must accompany this conclusion.

First, the TAA and SAA test windows are not identical in length. The 38-month TAA window is entirely contained within the longer 76-month SAA window and corresponds to a sub-period that, based on the SAA results, appears to have been particularly favourable for equity-oriented strategies. Some portion of the TAA outperformance may therefore reflect a more benign market environment rather than a genuine structural advantage of the Black-Litterman framework.

Second, and more fundamentally, the views incorporated in the TAA backtest were constructed ex post, with full knowledge of realised returns. The performance statistics should therefore be interpreted as an upper bound on what an accurate GMR team could achieve, rather than as a realistic estimate of live TAA performance. In a genuine real-time implementation, view accuracy would be imperfect, and the performance gap between TAA and SAA would be correspondingly narrower, and potentially negative if views are systematically miscalibrated.

7 Conclusion

The SAA component demonstrates a well-behaved and monotone risk–return gradient across the five risk profiles, confirming that the macro and relative constraints succeed in translating client risk preferences into structurally coherent allocations. Its principal strength lies in transparency and operational simplicity: once estimated on the training sample, the resampled weights require no discretionary input and are directly defensible to regulators and clients alike. Its main limitation is the sensitivity of the optimised weights to in-sample return estimates, which can diverge materially from out-of-sample realisations even after Michaud resampling, with the effect being most pronounced at the conservative end of the spectrum where all three least aggressive profiles fail to meet their annualised return targets. The Black-Litterman component addresses this limitation by replacing the raw sample mean with a Bayesian posterior that partially shrinks toward an equilibrium prior, producing more stable and diversified allocations. The out-of-sample TAA results, characterised by higher returns, substantially compressed volatility, near-uniform maximum drawdowns, and Sharpe ratios ranging from 0.92 to 2.13, illustrate the potential of the framework when research views are accurate. Its structural weakness is the dependence on the quality of GMR views: in the present backtest, views were selected ex post and therefore represent an upper bound on achievable performance rather than a realistic estimate of live alpha generation.

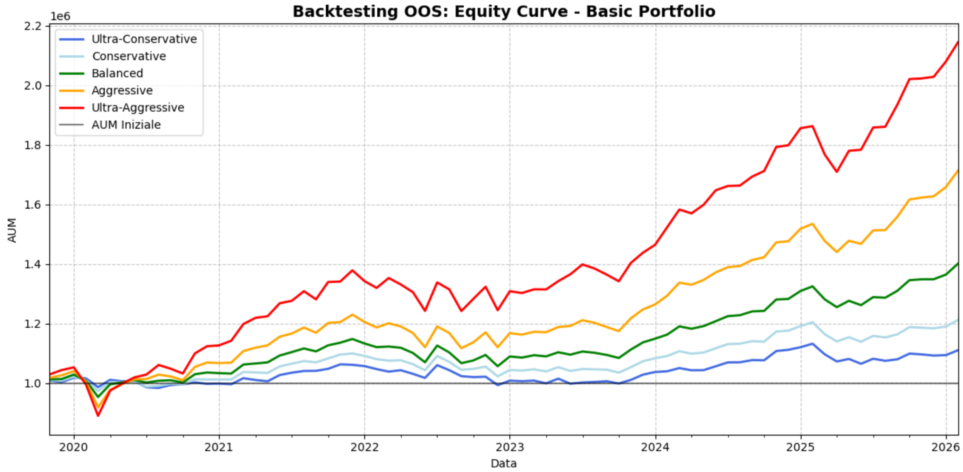

Figure 2: Backtesting Basic Portfolio for each Risk Profile.

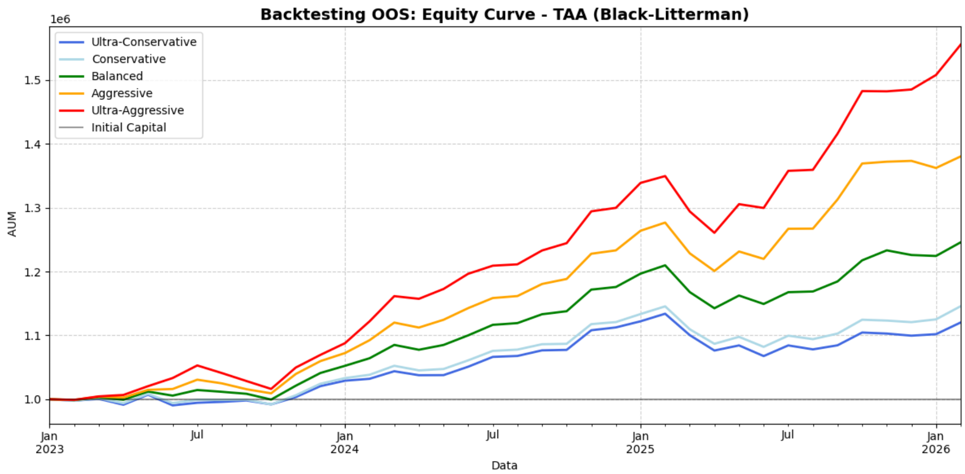

Figure 3: Backtesting BL portfolio for each Risk Profile.